StatML CDT: Selective Inference

Daniel G. Rasines and Alastair Young

Imperial College London

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Key references

- Zhang, Khalili, Asgharian (2022). "Post-model-selection inference in linear regression models: an integrated review", *Statistics Surveys*.
- Kuchibhotla, Kolassa, Kuffner (2022). "Post-selection inference", *Annual Review of Statistics and Its Application*.

Day 1

1. *Introduction*
2. *Simultaneous inference with a fixed design*
   - Berk, Brown, Buja, Zhang, Zhao (2013). "Valid post-selection inference", *Annals of Statistics*.
   - Bachoc, Preinerstorfer, Steinberg (2020). "Uniformly valid confidence intervals post-model-selection", *Annals of Statistics*.
   - Zrnic, Jordan (2022). "Post-selection inference via algorithmic stability", *arXiv:2011.09462*.
3. *Simultaneous inference with a random design*
   - Kuchibhotla, Brown, Buja, Cai, George, Zhao (2020). "Valid post-selection inference in model-free linear regression", *Annals of Statistics*.
   - Rinaldo, Wasserman, G'Sell (2019). "Bootstrapping and sample splitting for high-dimensional, assumption-free inference", *Annals of Statistics*.

Motivation

The **classical approach** to statistical inference assumes that all the **models to fit** and all the **inferential objectives** are **fixed prior to the data analysis**.

For example, a common regression problem assumes observation of a vector $Y \sim N(X\beta, \sigma^2 I)$ and seeks inference for β.

However, this is **not how statistics operates in practice**.

Motivation

Typically, the practitioner **interacts with the data** in order to select a suitable model to fit and/or a set of relevant inferential questions to address—the **selection stage**.

Such data exploration allows the practitioner to **focus only on the most relevant aspects** of the data-generating process.

BUT it invalidates the assumptions of a fixed model or inferential objectives, leading to a (possible) loss of the inferential guarantees indicated by classical theory.

Motivation

The further **sampling variability** introduced in the pre-analysis stage often leads to:

- Overstatement of statistical significance (exaggerated p-values).
- Confidence intervals with low coverage.
- Overestimation of effect sizes.
- Underestimation of variances.

Motivation

Selection effects are **regularly overlooked in statistical practice**, and are often cited as one of the main causes of the replicability crisis in science.

Famously, Breiman (1992) referred to this issue as a "quiet scandal in the statistical community".

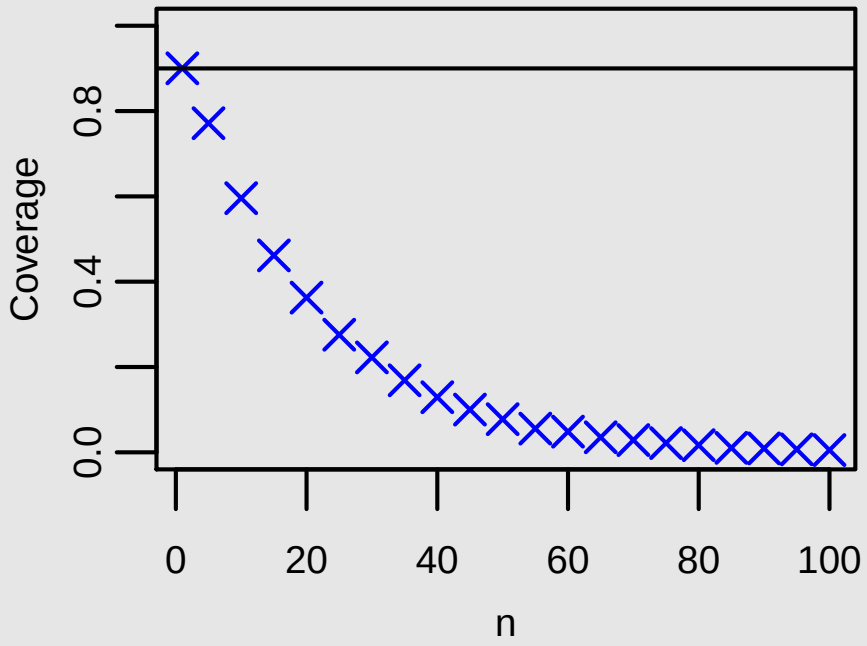
$\rightarrow$ The goal of **selective inference** is to **restore validity of inference** after selection.

Example: inference on winners

Let $Y_i \sim N(\theta_i, 1)$ independently for $i = 1, \ldots, n$, and suppose a confidence interval is required for the mean of the largest observation, $\theta_I(Y)$, where

$$I(Y) = \arg\max_{i=1,\ldots,n} Y_i.$$

Figure: Coverage of conventional 90% confidence intervals $[y_{I(Y)} - 1.64, y_{I(Y)} + 1.64]$ for the case $\theta_1 = \ldots = \theta_n$ as $n \to \infty$.

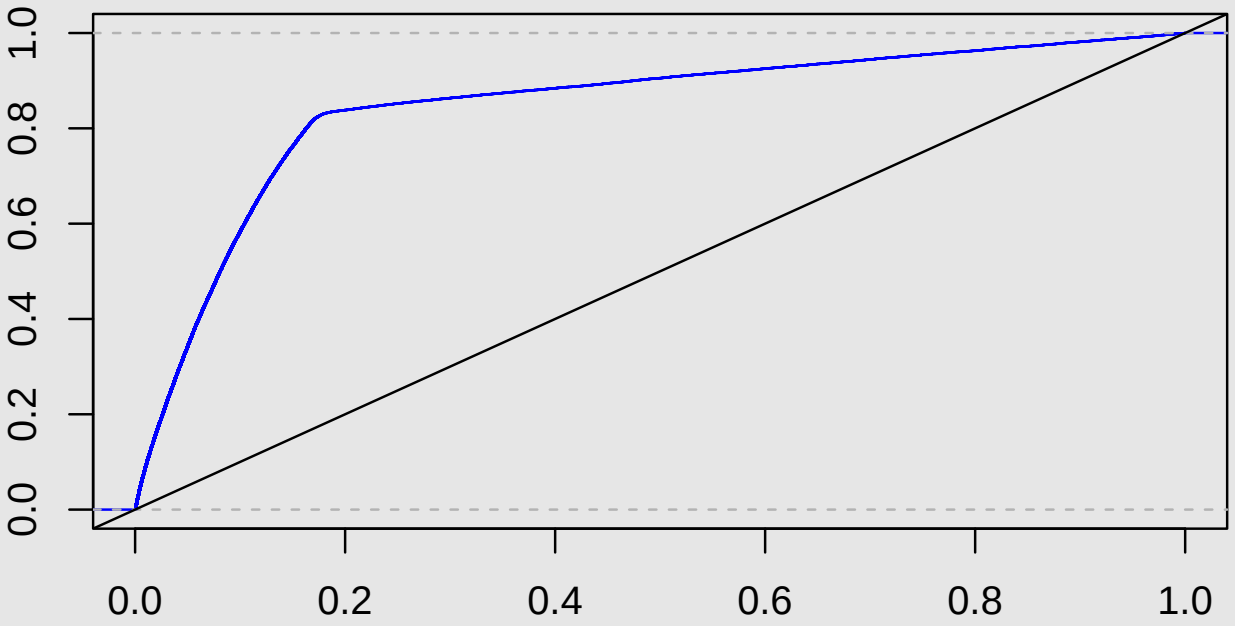


Example: inference after model selection

$Y \sim N(\mathbf{0}, I_{80})$ and $X \in \mathbb{R}^{80\times 20}$ with $N(0,1)$ entries independent of Y.

A forward stepwise algorithm minimising the AIC is used to select a linear submodel, and a marginal p-value is computed to test significance for each variable in the selected model.

Figure: CDF of the p-values.



$\sim$ **35% of selected coefficients deemed significant** at a 0.05 level, even though none of them are!

Approaches

Selective inference has a long and rich history.

Many approaches have been advocated to estimate/control selection effects, such as

- Simultaneous inference (Bonferroni, BH, ...).
- Bayesian methods: model averaging, empirical Bayes.
- Bootstrap.
- Differential privacy.

The goal of this course is to give an overview of recent proposals and describe how they fit within the general framework.

Framework

Data matrix denoted by $[Y, X]$, where

- $Y = (Y_1, \ldots, Y_n)^T \in \mathbb{R}^n$ is a quantitative response vector.
- $X = [X_1, \ldots, X_p] \in \mathbb{R}^{n \times p}$ is a $n \times p$ **fixed** design matrix, containing observations of p covariates.

In general (provided finite expectation), one can write

$$Y = \mu + \varepsilon, \text{ where } \mu = \mathbb{E}[Y], \quad \varepsilon = Y - \mu. \tag{1}$$

$\rightarrow$ Often μ is assumed to belong to a parametric class, such as the linear model $\text{span}(X) = \{\mu = X\beta \colon \beta \in \mathbb{R}^p\}$, and inference is sought for the corresponding parameter β.

Framework

If $p \sim n$ or $p > n$, it is common to seek a smaller model due to lack of identifiability and interpretability of the larger model.

We will denote by an index set $M = \{i_1, \ldots, i_m\} \subseteq \{1, \ldots, p\}$ a **linear submodel** containing only the covariates in M, where $m = |M|$.

Namely, if $X_M = [X_{i_1}, \ldots, X_{i_m}]$ denotes the submatrix of X that contains only the covariates in M, the linear submodel corresponding to M posits

$$Y = X_M \beta^M + \varepsilon, \quad \beta^M \in \mathbb{R}^m. \tag{2}$$

The projection parameter

In selective inference one normally treats **models as approximations** of a (potentially very complex) true underlying distribution.

This aligns more naturally with the model-selection framework and is considerably more realistic.

Even if $\mu \notin \text{span}(X_M) = \{X_M b \colon b \in \mathbb{R}^m\}$ for some M, one can still define a meaningful model-dependent parameter, the **projection parameter**:

$$\beta_M = \underset{b \in \mathbb{R}^m}{\arg\min}\, \mathbb{E}\|Y - X_M b\|^2 = (X_M^T X_M)^{-1} X_M^T \mu = X_M^\dagger \mu \in \mathbb{R}^m, \qquad (3)$$

for all M with $m < n$ and $\text{rank}(X_M) = m$, where $A^\dagger := (A^T A)^{-1} A^T$.

$\rightarrow$ In words, the projection parameter is the **best linear predictor** of μ in M with respect to the squared error loss.

The projection parameter

The entries of the projection parameter, denoted by β_{jM}, $j \in M$, have a different interpretation to the conventional regression coefficients:

$\rightarrow$ If $\mu = X_M \beta^M$, β_j^M is the average difference in the response for a unit difference in X_{i_j}, ceteris paribus in the model M.

$\rightarrow$ In general, β_{jM} is the average difference in the response **approximated in the submodel** M.

Naturally, if the selected model is in fact correct, then $\beta_M = \beta^M$ and the usual interpretation holds.

Interpretation of the projection parameter

For $j \in M$, let X_{jM} be the corresponding column of X_M, and define

$$r_{jM} = \left\{ I_n - X_{M\setminus\{j\}} X^{\dagger}_{M\setminus\{j\}} \right\} X_{jM}, \tag{4}$$

the residual vector of the regression of X_{jM} on the other predictors in M.

We can rewrite the j coefficient of β_M as

$$\beta_{jM} = \frac{1}{\|r_{jM}\|^2} r_{jM}^T \mu. \tag{5}$$

$\rightarrow$ β_{jM} is a **measure the relevance** of X_{jM} once we have adjusted for the other covariates in M.

Interpretation of the projection parameter

More specifically, denote by $P_M = X_M X_M^\dagger$ the projection matrix onto $\text{span}(X_M)$.

We can decompose it as

$$P_M = P_{M\setminus\{j\}} + (r_{jM}^T X_{jM}) r_{jM} r_{jM}^T. \tag{6}$$

$\rightarrow$ The **null hypothesis** $H_0 \colon \beta_{jM} = 0$ is equivalent to

$$P_M \mu = P_{M\setminus\{j\}} \mu, \tag{7}$$

i.e. it tests whether addition of X_{jM} improves the linear fit of $M \setminus \{j\}$.

PoSI

PoSI (**Po**st **S**election **I**nference) is a framework for selective inference for projection parameters with

- Finite sample guarantees for any dimension (n, p).
- Universal validity over (virtually) all model selection procedures.

Seminal work; introduced ideas which constitute the basis of much of the contemporary work on selective inference.

Crucially, it requires very **strong distributional assumptions**.

Extensions with less restrictive assumptions will be considered later.

PoSI

Assume $Y \sim N(\mu, \sigma^2 I_n)$, with $\mu \in \mathbb{R}^n$ and $\sigma^2 > 0$.

For a **fixed** submodel M, a natural estimator of β_M is

$$\hat{\beta}_M = X_M^\dagger Y \sim N(\beta_M, \sigma^2 (X_M^T X_M)^{-1}). \tag{8}$$

If a suitable estimate of σ^2 is available, confidence sets and testing procedures for β_M are provided by classical normal theory.

PoSI requires an estimator $\hat{\sigma}$ which is **independent** of all estimates $\hat{\beta}_{jM}$, and such that

$$\hat{\sigma}^2 \sim \sigma^2 \frac{\chi_r^2}{r}. \tag{9}$$

We write $r = \infty$ if σ^2 is known.

PoSI

Distributional requirements constitute a major practical limitation of the approach.

However, they are only needed for finite-sample validity; asymptotic guarantees can be derived under much weaker conditions, as we shall see.

A valid variance estimator will be available if $p < n$ and the full model $Y \sim N(X\beta, \sigma^2 I_n)$ is (at least approximately) valid.

PoSI

For a fixed model M, inference is straightforward.

Introduce the t-values relative to submodel M:

$$t_{jM} = \frac{\beta_{jM} - \hat{\beta}_{jM}}{[(X_M^T X_M)^{-1}]_{jj}^{1/2} \hat{\sigma}} \sim t_r, \quad j \in M. \tag{10}$$

A valid $1 - \alpha$ valid confidence interval for β_{jM} is given by

$$CI_{jM}(K) = [\hat{\beta}_{jM} \pm K[(X_M^T X_M)^{-1}]_{jj}^{1/2} \hat{\sigma}], \quad K = t_{r,1-\alpha/2}. \tag{11}$$

PoSI

In the selective context, however, the submodel M is allowed to depend on the data, i.e. it is **random**, which invalidates the previous CI.

Write a random submodel as $\hat{M} \equiv \hat{M}(Y) \subseteq \{1, \ldots, p\}$.

Think of $\hat{M}$ as the result of some variable-selection procedure such as LASSO, stepwise regression, visual diagnostics, etc.

In principle, $\hat{M} \colon \mathbb{R}^n \to \mathcal{M}_{\text{all}}$ can be any *measurable* map, where

$$\mathcal{M}_{\text{all}} = \{M \subseteq \{1, \ldots, p\} \colon \text{rank}(X_M) = |M|\}, \tag{12}$$

i.e. the projection parameter needs to be identifiable.

PoSI

Associated with a random model $\hat{M}$ there is a random projection parameter $\beta_{\hat{M}} = X_{\hat{M}}^{\dagger}\mu$, which constitutes the **moving target of inference**.

Note that

- $\beta_{\hat{M}}$ has random dimension $|\hat{M}|$.
- For a fixed j, it might not be the case that $j \in \hat{M}$.
- Conditionally on $j \in \hat{M}$, the parameter $\beta_{\hat{M}}$ is random.

Furthermore, the corresponding estimator

$$\hat{\beta}_{\hat{M}} = X_{\hat{M}}^{\dagger} Y \tag{13}$$

is **not normally distributed** due to the extra variability from the selection step, so classical normal theory does not apply.

PoSI

Question: Given the stochastic nature of the parameter, what constitutes a valid inferential procedure this setting?

In the fixed-M setting, the $1-\alpha$ confidence interval $CI_{jM}(K)$ for β_{jM} satisfies

$$P\left(\beta_{jM} \in CI_{jM}(K)\right) = 1-\alpha. \tag{14}$$

In the random-model setting, PoSI seeks a value of K such that

$$P\left(\beta_{j\hat{M}} \in CI_{j\hat{M}}(K)\ \forall\ j \in \hat{M}\right) \geq 1-\alpha, \tag{15}$$

for **any** selection procedure $\hat{M}$.

$\rightarrow$ K is called the **PoSI constant**.

PoSI

Key aspects of this approach:

- **Universality**: CIs are valid **regardless of the selection procedure**, even if this involves subjective and informal decisions; the practitioner is even allowed to change their mind and report a different model post-hoc.
- Intervals tend to be very **conservative** as a result: the actual coverage can be well above the nominal one for some selection rules.
- However, **there exists a selection procedure that requires full protection**: unless there is a strong reason for discarding certain ill-behaved selection rules, PoSI is optimal.
- It provides only **unconditional** guarantees (more on this later).
- Implemented via simulation (computationally demanding if $p \approx 20$).

PoSI: restricted model space

The conservative nature of PoSI can be partially alleviated under the assumption that not all models in $\mathcal{M}_{\text{all}}$ are being searched.

In many applications, there is **a priori knowledge about the set of plausible selected models**, e.g.

- A subset of the covariates is forced into the model (e.g. an intercept).
- There is a size restriction on the model: $|M| \leq k$ (*sparsity*).
- Hierarchical restrictions: polynomial regression, interactions, etc.

PoSI: restricted model space

If such assumption can be made, we denote by $\mathcal{M} \subseteq \mathcal{M}_{\text{all}}$ the *pre-specified* set of allowed models.

With sufficiently strong restrictions on $\mathcal{M}$ (particularly the sparsity one), the PoSI approach becomes **computationally manageable for large** p.

Further reduction can be achieved by discarding variables ignoring the response, e.g. if there is collinearity.

PoSI: the selection-adjusted constant

The PoSI constant K is formally defined as

$$K(X, \mathcal{M}, \alpha, r) = \min \left\{ K > 0 \colon \mathbb{P} \left(\max_{M \in \mathcal{M}} \max_{j \in M} |t_{jM}| \leq K \right) \geq 1 - \alpha \right\}, \tag{16}$$

where, recall,

$$t_{jM} = \frac{\beta_{jM} - \hat{\beta}_{jM}}{[(X_M^T X_M)^{-1}]_{jj}^{1/2} \hat{\sigma}} = \frac{e_j^T X_M^{\dagger} (Y - \mu)}{[(X_M^T X_M)^{-1}]_{jj}^{1/2} \hat{\sigma}}. \tag{17}$$

$T = \max_{M \in \mathcal{M}} \max_{j \in M} |t_{jM}|$ is **distribution constant**, so K is computable.

PoSI: proof of coverage control

For any measurable model-selection procedure $\hat{M} : \mathbb{R}^n \to \mathcal{M}$, we have the trivial bound

$$\max_{j \in \hat{M}} |t_{j\hat{M}}| \leq \max_{M \in \mathcal{M}} \max_{j \in M} |t_{jM}|. \tag{18}$$

Thus,

$$\mathbb{P}\left(\max_{j \in \hat{M}} |t_{j\hat{M}}| \leq K\right) \geq \mathbb{P}\left(\max_{M \in \mathcal{M}} \max_{j \in M} |t_{jM}| \leq K\right) \geq 1 - \alpha, \tag{19}$$

where $K = K(X, \mathcal{M}, \alpha, r)$.

PoSI: achievement of nominal coverage

Among all possible model-selection procedures, there is one for which the nominal covarage is achieved, i.e. for which the PoSI constant is **sharp**.

It is the **significance-hunting procedure**, which seeks the model with the most significant observed effect:

$$\hat{M}^*(Y) = \arg\max_{M \in \mathcal{M}} \max_{j \in M} |t_{jM}|. \tag{20}$$

Note that $\hat{M}^*$ depends on μ and is therefore **not actionable**.

However, protection against it is still required as it is a valid procedure.

PoSI: computation of the PoSI constant

Closed-form expressions for K are **not available**.

Brute force Monte Carlo needed to approximate the $1-\alpha$ quantile of

$$T = \max\{|t_{jM}| \colon M \in \mathcal{M}, j \in M\}. \tag{21}$$

Independent copies of T can be obtained by sampling $Z \sim N(0, I_n)$ and $S^2 \sim \chi^2_r$ independently and setting

$$t^*_{jM} = \frac{e_j^T X_M^\dagger Z}{[(X_M^T X_M)^{-1}]_{jj}^{1/2} S}, \quad T^* = \max\{|t^*_{jM}| \colon M \in \mathcal{M}, j \in M\}. \tag{22}$$

This algorithm is implemented in the R package `PoSI` (Buja, Zhang, Zhang, 2020).

PoSI: computation of the PoSI constant

Note:

- Computations are specific to the design X.
- If $\text{rank}(X) = p$ and $\mathcal{M} = \mathcal{M}_{\text{all}}$, need to evaluate $p2^{p-1}$ t-values, which becomes infeasible when p exceeds ~ 15 (note that there are 2^{p-1} models containing each covariate).
- Universal bounds for K, which are conservative but do not require simulations, can be used in high-dimensional problems.

PoSI: one primary predictor

Sometimes the analysis is centred on a predictor of interest, X_j, while the other predictors in M act as controls, so that

- The submodel space is $\mathcal{M}_j = \{M \in \mathcal{M}_{\text{all}} : j \in M\}$.
- Only the t-statistic associated with X_j is relevant.

In this context, the PoSI constant is defined differently:

$$K_j(X, \mathcal{M}, \alpha, r) = \min \left\{ K > 0 : \mathbb{P}\left(\max_{M \in \mathcal{M}_j} |t_{jM}| \leq K \right) \geq 1 - \alpha \right\}. \quad (23)$$

As in the unrestricted case, exact coverage with this constant is achieved by the significance-hunting procedure

$$\hat{M}_j^*(Y) = \arg\max_{M \in \mathcal{M}_j} |t_{jM}|. \quad (24)$$

PoSI: Scheffé bound

PoSI provides simultaneous inference for up to $p2^{p-1}$ linear contrasts β_{jM}.

Scheffé's method provides **simultaneous protection for all linear combinations** without all the computational burden.

Write

$$t_x = \frac{(Y-\mu)^T x}{\hat{\sigma}\|x\|}, \quad x \in \text{span}(X) \setminus \{\mathbf{0}\}. \tag{25}$$

Recall that for $x \propto r_{jM} \in \text{span}(X)$, $t_x = t_{jM}$, so simultaneous inference for all the directions in the column space of X is an overkill for our problem.

PoSI: Scheffé bound

Scheffé's constant, explicitly given by $K_S(\alpha, d, r) = \sqrt{dF_{d,r,1-\alpha}}$, with $d = \dim\{\text{span}(X)\}$, satisfies

$$\mathbb{P}\left(\sup_{x \in \text{span}(X)} |t_x| \leq K_S\right) = 1 - \alpha, \tag{26}$$

and thus it provides **valid selective intervals regardless of the design**.

Naturally $K \leq K_S$ for any X, and often the **difference is substantial**, but provides a valid solution when simulation of PoSI constant is too costly.

PoSI: size of the PoSI constant

Asymptotic bounds for $n \geq p$, $p \to \infty$ and $\mathcal{M} = \mathcal{M}_{\text{all}}$:

- **Lower bound**: $K = \Omega(\sqrt{\log p})$, achieved by orthogonal designs.
- **Upper bound**: $K = O(\sqrt{p})$, achieved by equicorrelated designs.

$\to$ For orthogonal X, $\beta_{jM} \equiv \beta_j$ for all (j, M), so only p directions need to be covered.

$\to$ Large range $(\sqrt{\log p}, \sqrt{p})$ suggests strong dependence on X.

$\to$ Scheffé constant has $K_S \sim \sqrt{p}$, so its optimality is case-dependent.

For **sparse model spaces**, $\mathcal{M}_s = \{M \subseteq \{1, \ldots, p\} \colon |M| \leq s\}$,

$$K = O\left(\sqrt{s \log(p/s)}\right). \tag{27}$$

PoSI: other universal bounds

A general upper bound for K is given by

$$Q_T\{g(\mathcal{M}, X), r, 1-\alpha/2\} \leq \frac{g(\mathcal{M}, X) + A}{1 - Br^{-1/2}} \tag{28}$$

for some known constants $A, B > 0$, where $Q_T(x, r, \alpha)$ is the α quantile of a non-central t distribution with r degrees of freedom and non-centrality parameter x, and

$$g(\mathcal{M}, X) = \mathbb{E}\left[\max_{M \in \mathcal{M}} \max_{j \in M} |w_{jM}^T Z|\right], Z \sim N(0, I_n), w_{jM}^T = e_{jM}^T X_M^\dagger / \|e_{jM}^T X_M^\dagger\|. \tag{29}$$

Two important cases:

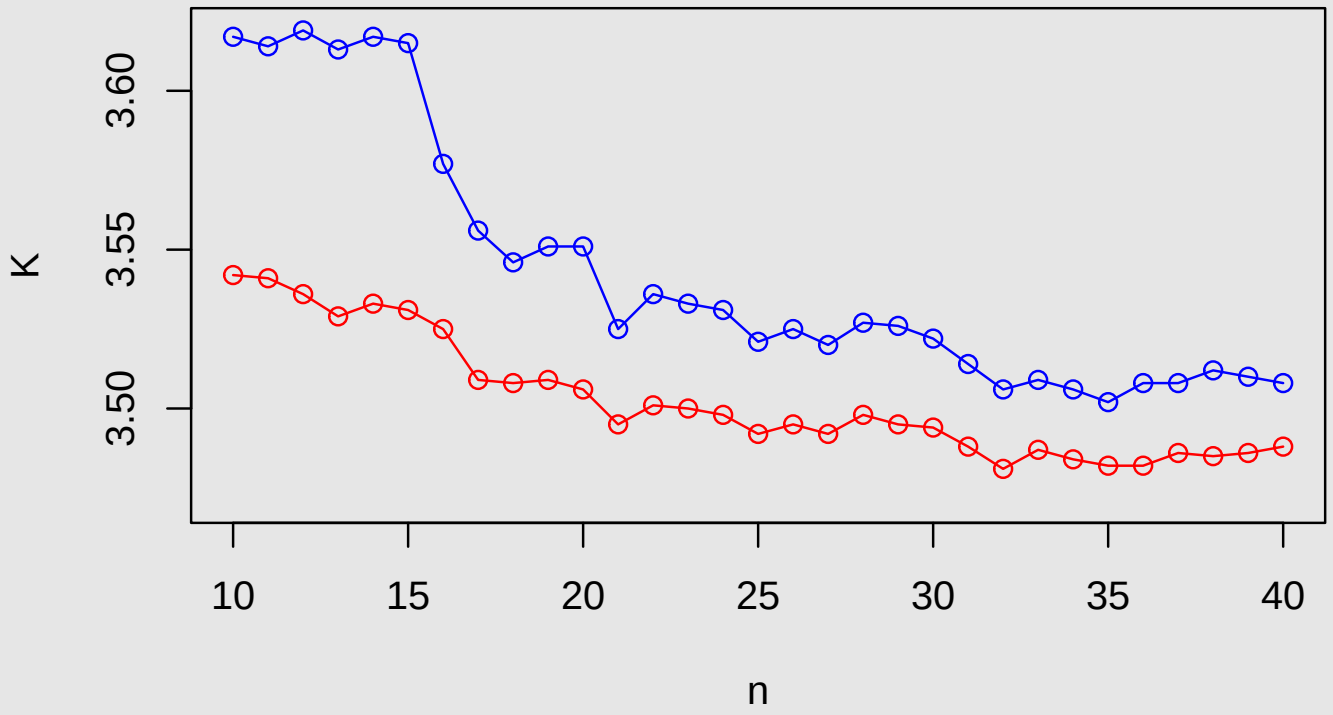
- Orthogonal designs: $g(\mathcal{M}, X) = \sqrt{2\log(2p)}$.
- Sparse models, $\mathcal{M}_s = \{M \colon |M| \leq s\}$: $g(\mathcal{M}, X) = \sqrt{2s\log(6p/s)}$.

Further refinements are available in the literature combining these two cases under the **Restricted Isometry Property**.

PoSI: size of the PoSI constant (empirical)

Set $p = 10$, $n \in \{10, \ldots, 40\}$, $r = \infty$ and $\alpha = 0.05$; predictors generated as Gaussian vectors with covariance $\Sigma_{ii} = 1$, $\Sigma_{ij} = 0.5$; Nsim $= 5 \times 10^4$.

Figure: PoSI constants for $\mathcal{M} = \mathcal{M}_{\text{all}}$ and $\mathcal{M} = \{M \colon |M| \leq 5\}$.



$\rightarrow$ The Scheffé constant for this problem is **4.28**.

PoSI: coverage

Empirical coverages of 95% CIs over 5000 samples; setting as before.

Data generated as $Y = X\beta + N(\mathbf{0}, I_n)$ under two parameters:

- $\beta^{(1)} = \mathbf{0}$.
- $\beta^{(2)} = (1, 1, -1, -1, 0, \ldots, 0)^T$.

Variable-selection rules:

- Lasso with cross-validation.
- Screening: a predictor is selected iff its significance p-value in the full linear model $\mu = X\beta$ is below 0.05.

$\mathbf{p = 10, n = 30}$

		Lasso	Screen.
$\beta^{(1)}$	PoSI	97.6	97.2
	Scheffé	99.8	99.8
	Unadj.	63.6	62.5
$\beta^{(2)}$	PoSI	99.5	99.5
	Scheffé	99.9	100
	Unadj.	93.4	92.8

$\mathbf{p = 10, n = 1000}$

	Lasso	Screen.
PoSI	95.7	95.4
Scheffé	99.7	99.8
Unadj.	50.5	50.4
PoSI	99.2	99.1
Scheffé	100	99.9
Unadj.	92.5	88.7

PoSI: extensions

The original PoSI framework has some practical limitations, most notably the restrictive distributional assumptions

$$Y \sim N(\mu, \sigma^2 I_n), \quad \hat{\sigma}^2 \sim \sigma^2 \frac{\chi_r^2}{r}. \tag{30}$$

Bachoc et al.[1] develop a more general framework that

- Provides **asymptotically valid** (*but* fixed-p) confidence intervals **without parametric assumptions** on the errors ε.
- Does **not require a consistent estimator** of σ.
- Is **applicable with other types of data (e.g. binary)**.

[1] Bachoc, Preinerstorfer, Steinberg (2020). "Uniformly valid confidence intervals post-model-selection". *Ann. Stat.*

Generalised PoSI: framework

- Data $Y = (Y_1, \ldots, Y_n)^T \sim \mathbb{P}_n$ has **independent but not necessarily identically distributed** components.
- $\mathbb{P}_n \in \mathbf{P}_n$, where $\mathbf{P}_n$ is a large non-parametric family of distributions.
- Statistician has a set of models $\mathcal{M}_n = \{M_{1,n}, \ldots, M_{d,n}\}$, possibly misspecified, where each $M_{j,n}$ is a set of distributions over $\mathcal{B}(\mathbb{R}^n)$.
- For each model $M \in \mathcal{M}_n$ there is a **prespecified parameter of interest**, $\theta_{M,n}(\mathbb{P}_n) \equiv \theta_{M,n}$, of dimension $m(M)$, and a corresponding estimator $\hat{\theta}_{M,n}$. Typically $\theta_{M,n}$ is a projection of $\mathbb{P}_n$ onto M.

Note: We can phrase the original PoSI problem as a special case of this framework with $\mathbb{P}_n = N(\mu_n, \sigma^2 I_n)$, $M = \{N(X_M \beta^M, \sigma^2 I_n) \colon \beta^M, \sigma^2\}$, and $\theta_{M,n} = X_M^\dagger \mu_n$.

Generalised PoSI: objective

For a fixed $\alpha \in (0,1)$, define a family of intervals for $\theta^{(j)}_{M,n}$,

$$\{CI^{(j)}_{1-\alpha,M} \colon M \in \mathcal{M}_n, 1 \leq j \leq m(M)\}, \tag{31}$$

satisfying

$$\liminf_{n\to\infty} \mathbb{P}_n \left(\theta^{(j)}_{M,n} \in CI^{(j)}_{1-\alpha,M} \text{ for all } 1 \leq j \leq m(M), M \in \mathcal{M}_n\right) \geq 1-\alpha. \tag{32}$$

It then follows that

$$\liminf_{n\to\infty} \mathbb{P}_n \left(\theta^{(j)}_{\hat{M}_n,n} \in CI^{(j)}_{1-\alpha,\hat{M}_n} \text{ for all } 1 \leq j \leq m(\hat{M}_n)\right) \geq 1-\alpha \tag{33}$$

for *any* model-selection procedure $\hat{M}_n \colon \mathbb{R}^n \to \mathcal{M}_n$.

Generalised PoSI: notation

Let $\theta_n = (\theta_{M_1,n}^T, \ldots, \theta_{M_d,n}^T)^T$ and $\hat{\theta}_n = (\hat{\theta}_{M_1,n}^T, \ldots, \hat{\theta}_{M_d,n}^T)^T$.

Assume regularity conditions such that

$$d\left\{\mathrm{diag}(V_n)^{\dagger/2}(\hat{\theta}_n - \theta_n), N(0, \mathrm{corr}(V_n))\right\} \to 0 \quad \text{as } n \to \infty. \tag{34}$$

for a sequence of covariances V_n, where $d\{\cdot\}$ is any distance metrising convergence in distribution, and

- $\mathrm{diag}(A)$ is the diagonal matrix sharing the diagonal with A.
- $A^{1/2}$ is the square root of a SPD matrix A.
- $A^{\dagger/2} = (A^\dagger)^{1/2}$.
- $\mathrm{corr}(A) = \mathrm{diag}(A)^{\dagger/2} A \, \mathrm{diag}(A)^{\dagger/2}$.

Generalised PoSI: notation

Define $K_{1-\alpha}(\Sigma)$ as the $1-\alpha$ quantile of $\|Z\|_\infty$, where $Z \sim N(0, \Sigma)$.

This new "PoSI constant" is based on a Gaussian distribution because the guarantees of this method are asymptotic for fixed p.

The original PoSI constant in the known-σ^2 case ($r = \infty$) can be written as $K_{1-\alpha}(\mathrm{corr}(\Gamma_X))$, where the $|M_i| \times |M_j|$ block of Γ_X is given by

$$X^\dagger_{M_i}(X^\dagger_{M_j})^T, \tag{35}$$

for $M_i, M_j \in \mathcal{M}$.

Generalised PoSI: consistent variance estimation

Theorem

*(**Bachoc et al., Theorem 2.3**) Let $\hat{V}_n$ be a consistent estimator of V_n. Under certain asymptotic conditions on $\hat{\theta}_n$ and $\hat{V}_n$, the intervals*

$$CI^{(j)}_{1-\alpha,M} = \hat{\theta}^{(j)}_{M,n} \pm \sqrt{[\hat{V}_n]_{\rho(M)+j}} K_{1-\alpha}(corr(\hat{V}_n)) \tag{36}$$

are asymptotically valid post-selection $1-\alpha$ confidence intervals for θ_{jM}, where for $M = M_{j,n}$, $\rho(M) = \sum_{l=1}^{j-1} m(M_{l,n})$, and $[A]_j$ is the j-th diagonal entry of A.

$\rightarrow$ Since guarantees are only aymptotic, the PoSI constant is computed from a multivariate Gaussian instead of a multivariate t distribution.

$\rightarrow$ In a fully non-parametric setting consistent estimation of the variance is rarely possible, so this construction is still of limited use.

Generalised PoSI: variance overestimation

Theorem

*(**Bachoc et al., Theorem 2.5**) Suppose there exists estimators $\hat{v}^2_{j,n}$ of $[V_n]_j$, and an estimator $\hat{K}_n$ of $K_{1-\alpha}(corr(V_n))$ such that*

$$\mathbb{P}\left(\frac{K_{1-\alpha}(corr(V_n))}{\hat{K}_n}\max_j\sqrt{\frac{[V_n]_j}{\hat{v}^2_{j,n}}} > 1+\varepsilon\right) \to 0 \text{ for all } \varepsilon > 0. \tag{37}$$

Then, under the same conditions as before, the intervals

$$CI^{(j)}_{1-\alpha,M} = \hat{\theta}^{(j)}_{M,n} \pm \sqrt{\hat{v}^2_{\rho(M)+j,n}}\hat{K}_n \tag{38}$$

are asymptotically valid post-selection $1-\alpha$ confidence intervals for θ_{jM}.

→ Estimators $\hat{v}^2_{j,n}$ are possible to construct in a variety of settings.

→ Widely applicable upper bounds for $K_{1-\alpha}(\text{corr}(V_n))$ are available.

Generalised PoSI: homoskedastic quantitative data

Consider a misspecified version of original PoSI, where

- The true distribution of Y is such that the entries are independent, have identical variance σ_n^2, and

$$\max_{i=1,\ldots,n} \left[\mathbb{E}\left(|Y_i - \mathbb{E}(Y_i)|^{2+\delta}\right)\right]^{\frac{2}{2+\delta}} \leq \tau\sigma_n^2 \tag{39}$$

  for some $\delta > 0$, $\tau \geq 1$.
- The set of working models $\mathcal{M}_n$ for Y contains homoskedastic linear models $\mathbb{E}(Y) = X_M\beta_M$, $M \subseteq \{1, \ldots, p\}$, with p fixed.

Asymptotically valid selective CIs for projection parameters are given by

$$CI_{1-\alpha,M}^{(j)} = \hat{\beta}_{M,n}^{(j)} \pm \sqrt{\hat{\sigma}_{M,n}^2[(X_M^T X_M)^{-1}]_{jj}}K_{1-\alpha}(\mathrm{corr}(\Gamma_X)); \tag{40}$$

$$\hat{\sigma}_{M,n}^2 = \frac{\|(I_n - P_M)Y\|^2}{n - |M|}. \tag{41}$$

Generalised PoSI: homoskedastic quantitative data

The resulting intervals are very similar to original PoSI intervals, but there are two key differences:

- The variance is estimated using the **residual sum of squares from the fit of the selected model**. This will in general overestimate the true variance, but adapts to misspecification.
- Since validity is guaranteed only asymptotically, the PoSI constant is computed from a multivariate Gaussian (rather than t) distribution.

This framework can be applied to heteroskedastic and binary data.

PoSI: example

Diabetes dataset of Hastie and Efron (2012).

442 patients, 10 covariates.

Response: quantitative measure of disease progression.

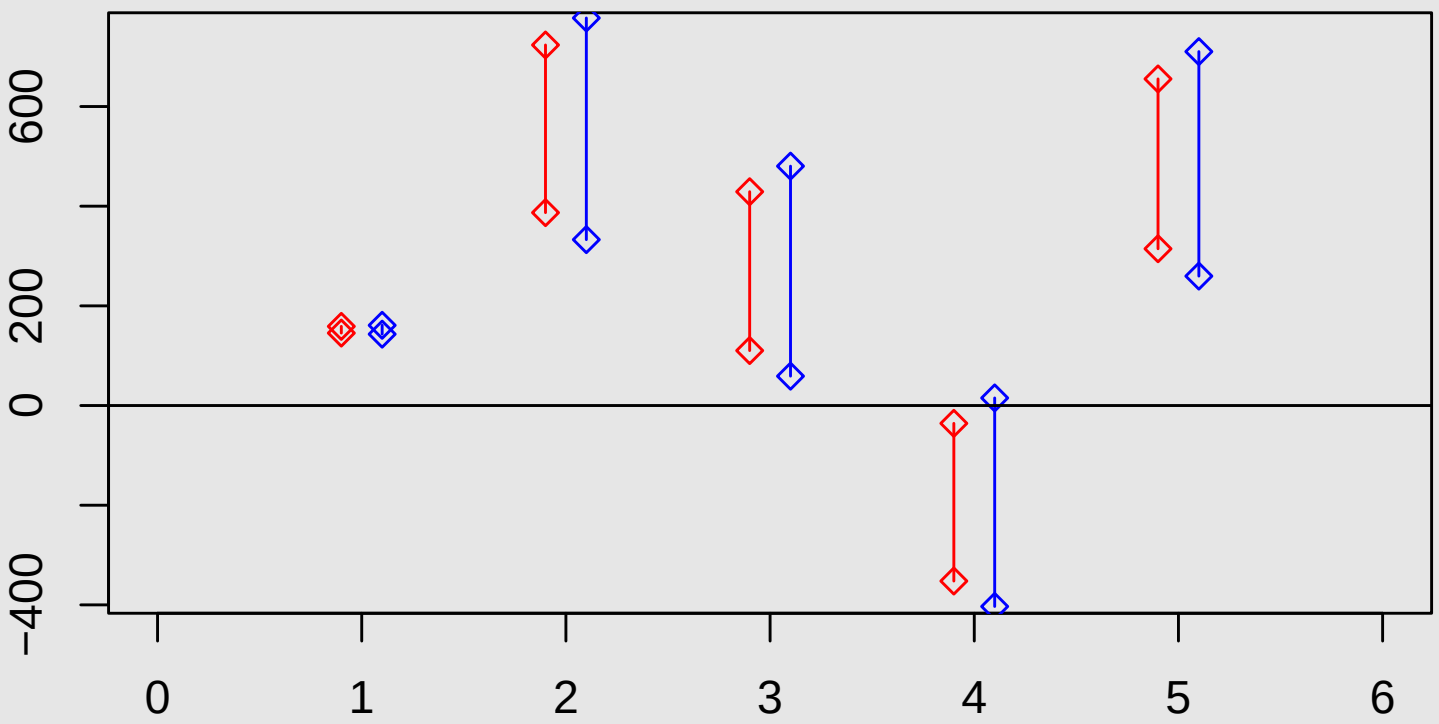
Covariates: age, sex, BMI, blood pressure, and 6 blood serum measurements.

We run the LASSO with penalty selected by cross-validation to identify the most significant predictors, obtaining:

`BMI MAP HDL LTG`

PoSI: example

Figure: Comparison of unadjusted intervals with PoSI intervals at 95% level.



→ HDL is deemed significant (within the selected model) if the selection step is ignored, but when the appropriate adjustment is made there is no ground for rejection.

Selective inference via stability

PoSI is necessarily **conservative** because it is required to protect against *any* selection rule, even those which make little sense, which also makes it **computationally expensive**.

To remedy this, in addition to restricting the model space, one can decrease the size of the PoSI confidence regions by limiting the set of allowed selection rules.

Motivated by ideas from differential privacy, a powerful restriction comes via the notion of **algorithmic stability**.

Loosely speaking, a selection rule is stable if it is **not very sensitive to the data**.

Idea: for a stable selection rule, the degree of dependence between model selection and inference is bounded, and therefore the selection bias can be controlled.

Selective inference via stability

Stability is a property of **randomised selection rules**.

The output of a randomised selection rule $\hat{M}$, $\hat{M}(y)$, is a *random variable* on the power set of $\{1, \ldots, p\}$.

In general, starting from a deterministic rule $\hat{M}\colon \mathbb{R}^n \to \mathcal{M}$, $\mathcal{M} \subseteq 2^{\{1,\ldots,p\}}$, a randomised version can be obtained by applying it to a *noisy* version of the data, such as $\tilde{Y} = Y + W$, where W is artificial noise from a known distribution generated by the statistician.

Conditionally on $Y = y$, $\hat{M}(y + W)$ is a random variable whose output is more or less similar to $\hat{M}(y)$ depending on the distribution of W.

In general, stability is a property of randomised algorithms (not necessarily model selection rules), $\mathcal{A}\colon \mathbb{R}^n \to \mathcal{S}$, where $\mathcal{S}$ is a set of random variables.

Selective inference via stability

Closeness between the output of two randomised algorithms is formalised via the concept of *indistinguishability* (aka *max-divergence*).

Definition

A random variable A is (η, τ)-**indistinguishable** from a random variable B, denoted $A \approx_{\eta,\tau} B$, if for all measurable sets E,

$$P(A \in E) \leq e^{\eta} P(B \in E) + \tau. \tag{42}$$

The parameters are $\eta > 0$ and $\tau \in [0, 1]$.

$\rightarrow$ In the context of differential privacy, η is a user-specified parameter that controls the trade-off between security and accuracy, and τ is the probability of having a security breach.

$\rightarrow$ In selective inference, we think of η as **balancing the information** allocated to selection and inference respectively, and of τ as being proportional to the **miscoverage probability** of a confidence set.

Selective inference via stability

With this notion in mind, we define a stable randomised algorithm as one for which, for most samples y, we can "guess" the distribution of $\mathcal{A}(y)$ (given y) knowing only the underlying distribution P but not the data y.

Definition

Let $\mathcal{A}$ be a randomised algorithm. We say it is (η, τ, ν)-**stable** with respect to a distribution P on $\mathbb{R}^n$ if there exists a random variable A_0, possibly depending on P, such that

$$P\left(y \in \mathbb{R}^n \colon \mathcal{A}(y) \approx_{\eta,\tau} A_0\right) \geq 1 - \nu. \tag{43}$$

When the algorithm is a model-selection rule, we write $\mathcal{A}$ and A_0 as $\hat{M}$ and M_0.

Selective inference via stability

Crucially, stability entails the existence of a random variable M_0, which can be thought of as an **oracle selection rule**, which is **independent of the data**, and therefore does not generate selection bias.

So, if we had access to M_0, we could use it instead and report classical confidence intervals for the selected parameter.

The following result makes this intuition rigorous.

Lemma

Let $\hat{M}$ be (η, τ, ν)-stable selection rule and M_0 be the corresponding oracle rule. Then

$$[Y, \hat{M}(Y)] \approx_{\eta,\tau+\nu} [Y, M_0], \tag{44}$$

with Y and M_0 independent.

Selective inference via stability

Consider a selective inference problem with data $Y \in \mathbb{R}^n$ and model-dependent targets of inference $\{\beta_M \colon M \in \mathcal{M}\}$.

Suppose that we can find a collection $\{\hat{R}_M^\alpha(\cdot) \colon M \in \mathcal{M}, \alpha \in (0,1)\}$ such that

$$P\left(\beta_M \in \hat{R}_M^\alpha(Y)\right) \geq 1 - \alpha \text{ for all } M \in \mathcal{M}, \alpha \in (0,1). \tag{45}$$

Theorem

Fix $\delta \in (0,1)$ and let $\hat{M}$ be a (η, τ, ν)-stable selection algorithm. Then

$$P\left\{\beta_{\hat{M}} \in \hat{R}_{\hat{M}}^{\delta e^{-\eta}}(Y)\right\} \geq 1 - (\delta + \tau + \nu). \tag{46}$$

$\rightarrow$ Under stability, we can effectively ignore selection and report classical intervals with a "selection-adjusted" nominal coverage.

For a given a (δ, τ, ν) and a desired coverage $1 - \alpha$, the adjusted nominal coverage needs to be $\tilde{\alpha} = (\alpha - \tau - \nu)e^{-\eta}$.

Selective inference via stability

Proof.

By the lemma,

$$\begin{aligned} P\left\{\beta_{\hat{M}} \notin \hat{R}_{\hat{M}}^{\delta e^{-\eta}}(Y)\right\} &\leq e^{\eta} P\left\{\beta_{M_0} \notin \hat{R}_{M_0}^{\delta e^{-\eta}}(Y)\right\} + \tau + \nu \\ &= e^{\eta} \mathbb{E}\left[P\left\{\beta_{M_0} \notin \hat{R}_{M_0}^{\delta e^{-\eta}}(Y) \mid M_0\right\}\right] + \tau + \nu. \end{aligned}$$

By construction,

$$P\left\{\beta_{M_0} \notin \hat{R}_{M_0}^{\delta e^{-\eta}}(Y) \mid M_0\right\} \leq \delta e^{-\eta}, \tag{47}$$

and therefore

$$P\left\{\beta_{\hat{M}} \notin \hat{R}_{\hat{M}}^{\delta e^{-\eta}}(Y)\right\} \leq e^{\eta} e^{-\eta} \delta + \tau + \nu = \delta + \tau + \nu. \tag{48}$$

□

Selective inference via stability

In the PoSI framework, where $Y \sim N(\mu, \sigma^2 I_n)$ and $\beta_M = X_M^\dagger \mu$, these confidence regions specialise to the joint confidence intervals

$$\left[\hat{\beta}_{j\hat{M}} \pm K_{\hat{M}}(\delta e^{-\eta})[(X_{\hat{M}}^T X_{\hat{M}})^{-1}]_{jj}^{1/2}\hat{\sigma}\right], \quad j = 1, \ldots, |\hat{M}| \tag{49}$$

where $\hat{\sigma}$ is a suitable estimator of σ, and $K_M(\alpha)$ is a model-dependent PoSI constant: the minimum value of K satisfying

$$P\left(\max_{1 \le j \le |M|} \left| \frac{\hat{\beta}_{jM} - \beta_{jM}}{[(X_M^T X_M)^{-1}]_{jj}^{1/2}\hat{\sigma}} \right| \le K\right) \ge 1 - \alpha. \tag{50}$$

Selective inference via stability

In many cases, stable selection rules can be derived from existing ones via addition of Laplace noise.

Let $Y \sim N(\mu, \sigma^2 I_n)$ for a known $\sigma > 0$, and suppose that the selection rule is a function of the randomised algorithm $\mathcal{A}(Y) = a^T Y + W$ for some fixed $a \in \mathbb{R}^n$, where

$$W \sim \text{Laplace}\left(\frac{z_{1-\nu/2}\sigma\|a\|}{\eta}\right), \tag{51}$$

This is, $\hat{M} = M \iff \mathcal{A}(Y) \in E_M$ for some $E_M \subseteq \mathbb{R}$.

We have that $\hat{M}$ is $(\eta, 0, \nu)$-stable.

Selective inference via stability

To see this, let $\mathcal{Y}_\nu = \{x \in \mathbb{R}^n \colon |a^T x - a^T \mu| \leq z_{1-\nu/2}\sigma\|a\|\}$, so that

$$P(\mathcal{Y}_\nu) = P(|N(0,1)| \leq z_{1-\nu/2}) = 1 - \nu. \tag{52}$$

Consider the oracle randomised algorithm $\mathcal{A}(\mu) = a^T\mu + W$, which uses the true unknown mean.

Since the density ratio of a Laplace(b) and a u+Laplace(b) distribution is upper bounded by $e^{|u|/b}$, we have, for all $x \in \mathcal{Y}_\nu$ and all measurable E,

$$\frac{P\{\mathcal{A}(x) \in E\}}{P\{\mathcal{A}(\mu) \in E\}} \leq e^\eta. \tag{53}$$

Therefore, $\mathcal{A}$ and any function of it are $(\eta, 0, \nu)$-stable.

Selective inference via stability

From a basic stable rule such as the previous one, more sophisticated stable rules can be derived using the following two properties.

Postprocessing:

If $\mathcal{A}$ is (η, τ, ν)-stable, then any composition $\mathcal{B} \circ \mathcal{A}$ is also (η, τ, ν)-stable.

For example, if we have a stable version of the lasso algorithm, with maximiser $\hat{\beta}$, the selection rule $\hat{M} = \{1 \leq j \leq p \colon \hat{\beta}_j \neq 0\}$ is also stable.

Composition:

Consider a sequence of algorithms $\mathcal{A}_t \colon \mathcal{S}_1 \times \cdots \times \mathcal{S}_{t-1} \times \mathbb{R}^n \to \mathcal{S}_t$ for $t = 1, \ldots, k$. If each algorithm is stable, their composition is also stable, and the stability parameters can be written as a function of the stability parameters of the $\mathcal{A}_t$'s.

In its most simple form, if each algorithm is $(\eta, 0, 0)$-stable, their composition is $(k\eta, 0, 0)$-stable.

For example, this can be applied to stepwise algorithms if each step is known to be stable.

Selective inference via stability

As an example, consider a stable version of the lasso algorithm.

For data $[Y, X] \in \mathbb{R}^n \times \mathbb{R}^{n \times p}$, the (standard) lasso estimator is the solution of

$$\hat{\beta} = \arg\min_{\beta} \frac{1}{2} \| Y - X\beta \|_2^2 \text{ s.t. } \|\beta\|_1 \leq C, \tag{54}$$

for a chosen L_1 restriction penalty $C > 0$.

The stable lasso is obtained by running a numerical optimisation routine for (54) adding Laplacian noise at each step.

Selective inference via stability

The tuning parameters of the algorithm are the *penalty* C, the *number of steps* k, and the *stability-inducing parameters* δ and η. Furthermore, let $\{e_i\}_{i=1}^p$ be the canonical basis of $\mathbb{R}^p$. For theoretical guarantees, we require $Y \sim N(\mu, \sigma^2 I_n)$ and assume we have an estimator $\hat{\sigma}$ of σ.

<u>Stable lasso</u>: Set $\beta^{(1)} = 0$. For $t = 1, \ldots, k$:

1. For all $\phi \in C \cdot \{\pm e_i\}_{i=1}^p$, sample

$$W_{t,\phi} \sim \text{Laplace}\left(\frac{4t_{r,1-\delta/(2d)} C \|X\|_{2,\infty}}{\eta n}\right), \|X\|_{2,\infty} = \max_i \|X_i\|_2. \tag{55}$$

2. For all $\phi \in C \cdot \{\pm e_i\}_{i=1}^p$, let $\alpha_\phi = -\frac{2}{n\hat{\sigma}} \phi^T X^T (Y - X\beta_t) + W_{t,\phi}$.
3. Set $\phi_t = \arg\min \alpha_\phi$.
4. Set $\beta^{(t+1)} = (1 - \Delta_t)\beta^{(t)} + \Delta_t \phi_t$, where $\Delta_t = 2/(t+1)$.

Report $\hat{\beta} = \beta^{(k+1)}$.

Selective inference via stability

This is essentially a randomised version of the classical Frank-Wolfe algorithm for the standard lasso which uses the stability of the additive Laplace noise as a building block. We have the following result.

Theorem

Assume that $Y \sim N(\mu, \sigma^2 I_n)$ and $\hat{\sigma}^2 \sim \sigma^2 \chi_r^2 / r$. The randomised lasso algorithm is both

- $(k\eta^2/2 + \sqrt{2k \log(1/\delta)}, \delta, \delta)$*-stable;*
- $(k\eta, 0, \delta)$*-stable.*

By the post-processing property, any model-selection procedure based on the output of this algorithm, such as

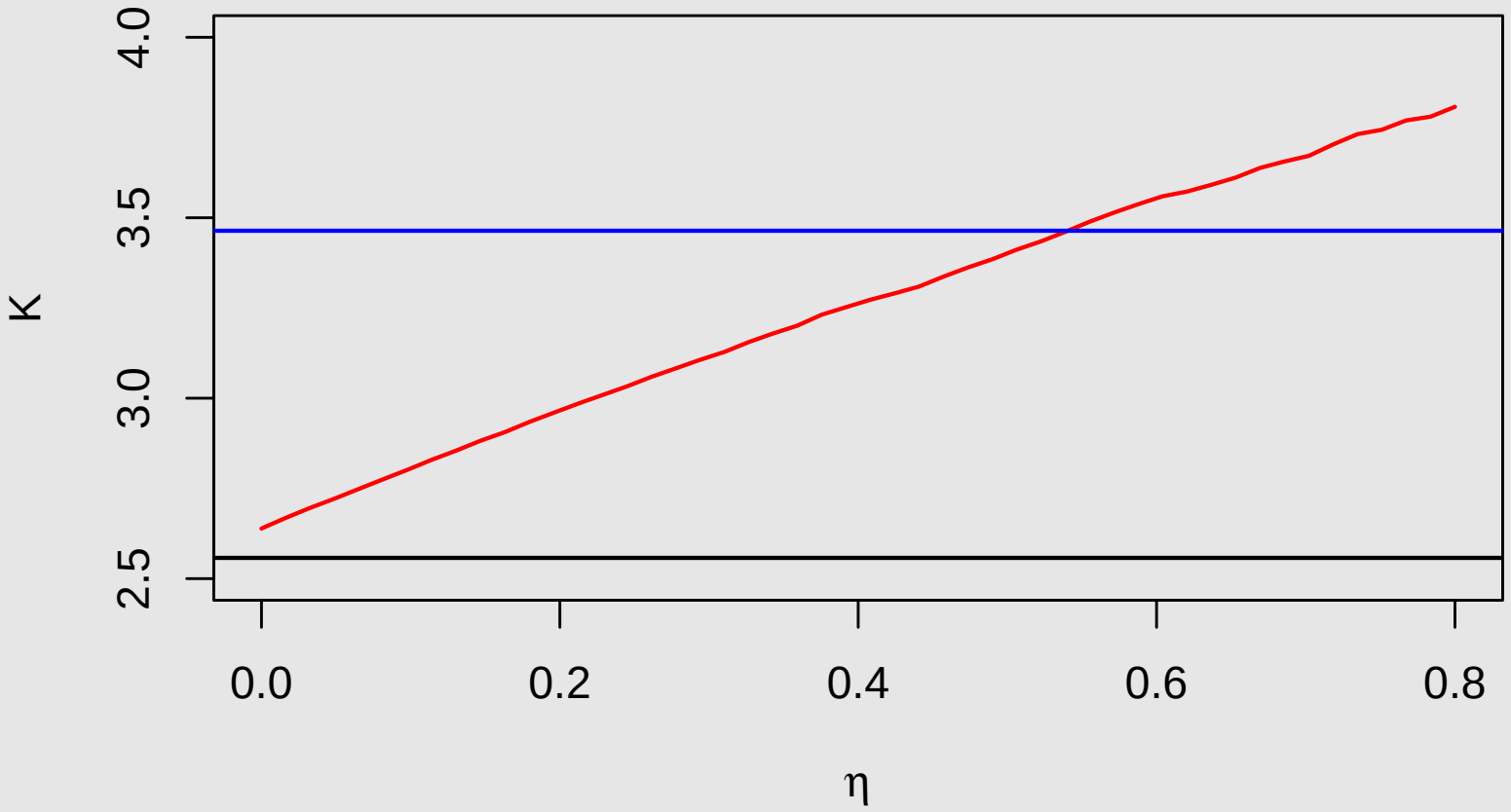
$$\hat{M} = \{j \in \{1, \ldots, p\} \colon \hat{\beta}_j \neq 0\}, \tag{56}$$

is stable.

$\rightarrow$ Similar results hold for stable versions of other well-established procedures such as marginal screening.

Selective inference via stability

Figure: Red: (optimal) multiplicative adjustment factor for the stable lasso in a setting with $n = 50$, $p = 10$, $|M| = 5$, $k = 5$, $\delta = 0.01$, and $0 \leq \eta \leq 0.8$; Blue: PoSI constant; Black: unadjusted constant.



Fixed vs. random design

So far we have assumed that the design X is **fixed**.

This makes modelling and computation easier.

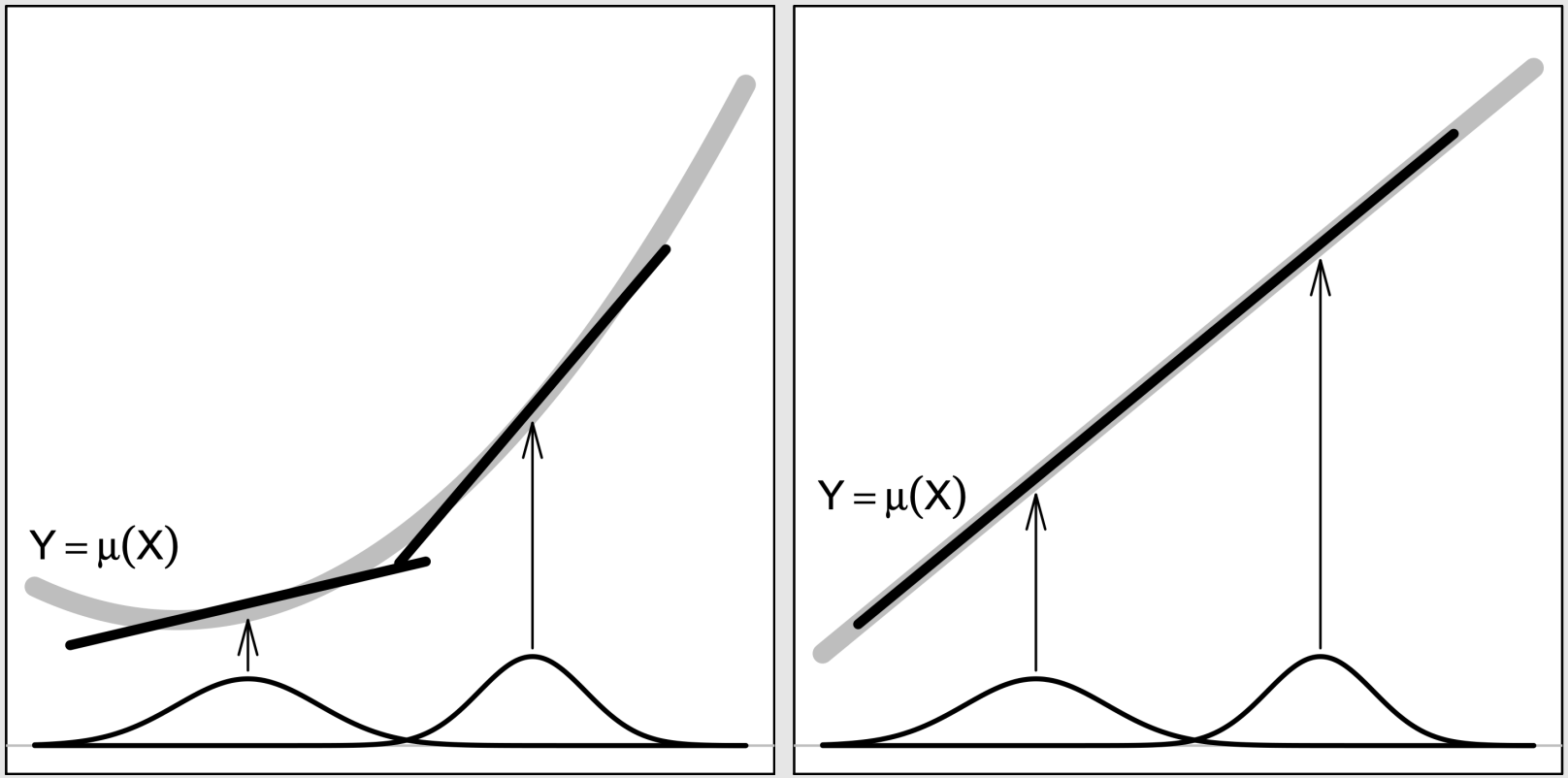
Conceptually, it can be justified in one of the following cases:

- The covariate values are fixed by the experimenter.
- Ancillarity argument: X is random but its distribution is independent of any parameter of interest. Then, by the *Conditionality Principle*, inference ought to be provided conditionally on the observed X.

The latter assumption is often sensible when the working model is well-specified, but is problematic under misspecification.

Fixed vs. random design

Figure: Buja et al. (2020). "Models as approximations I: consequences illustrated with linear regression".



$\rightarrow$ The best linear approximation of the true regression function $\mu(X) = \mathbb{E}[Y \mid X]$ **depends on the distribution of X**.

Fixed vs. random design

In view of this, some authors have approached the selective inference problem from a random-design perspective.

Two main proposals:

- UPoSI: reformulation of the original PoSI framework in the random-X setting.
- Data splitting: basing model selection and inference on independent subsamples of the data. Inferential step is thereby unaffected by selection, allowing usage of standard statistical techniques.

The inferential target

Let $(Y_i, X_i^T)^T \in \mathbb{R} \times \mathbb{R}^p$, $i = 1, \ldots, n$ be n samples of observations, **independent but not necessarily identically distributed**, and set $Y = (Y_1, \ldots, Y_n)^T$ and $X = [X_1, \ldots, X_p]^T$.

For a model $M \subseteq \{1, \ldots, p\}$ define the population and empirical risks of the squared loss:

$$R_n(\theta; M) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}\left[\{Y_i - X_i(M)^T\theta\}^2\right] \tag{57}$$

$$\hat{R}_n(\theta; M) = \frac{1}{n} \sum_{i=1}^{n} \{Y_i - X_i(M)^T\theta\}^2, \tag{58}$$

where $X_i(M) = (X_{ij})_{j \in M} \in \mathbb{R}^{|M|}$, and their respective minimisers

$$\beta_{n,M} = \arg\min_{\theta \in \mathbb{R}^{|M|}} R_n(\theta; M), \quad \hat{\beta}_{n,M} = \arg\min_{\theta \in \mathbb{R}^{|M|}} \hat{R}_n(\theta; M). \tag{59}$$

The inferential target

Define the full-model second order matrices and vectors

$$\Sigma_n = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[X_i X_i^T\right], \qquad \hat{\Sigma}_n = \frac{1}{n}\sum_{i=1}^{n} X_i X_i^T \tag{60}$$

$$\Gamma_n = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[X_i Y_i^T\right], \qquad \hat{\Gamma}_n = \frac{1}{n}\sum_{i=1}^{n} X_i Y_i^T \tag{61}$$

The analogous quantities for a given model M are defined with $X(M)$ instead of X. They are simply submatrices and subvectors of the full-model versions.

The least squares parameter and its estimator satisfy

$$\hat{\Sigma}_n(M)\hat{\beta}_{n,M} = \hat{\Gamma}_n(M) \text{ and } \Sigma_n(M)\beta_{n,M} = \Gamma_n(M). \tag{62}$$

The inferential target

Some remarks:

- $\beta_{n,M}$ is the best linear predictor of Y using $X(M)$ relative to the squared loss.
- The normal equations are written implicitly because the framework allows for non-unique minimisers. In such cases the corresponding confidence regions will contain a subspace of $\mathbb{R}^p$.
- For a fixed M,

$$\sqrt{n}\{\hat{\beta}_{n,M} - \beta_{n,M}\} \xrightarrow{d} N(0, V_M) \tag{63}$$

  for some matrix $V_M > 0$ under mild conditions.
- In general, for $M \subseteq M'$ it does not hold that $\beta_{n,M}$ is a subvector of $\beta_{n,M'}$ due to dependence among the columns of X.

The inferential target

As usual focus on inference via confidence regions.

Asymptotic normality lends itself for the construction of regions $\hat{R}_{n,M} \subseteq \mathbb{R}^{|M|}$ such that

$$\liminf_{n\to\infty} P\left(\beta_{n,M} \in \hat{R}_{n,M}\right) \geq 1 - \alpha. \tag{64}$$

UPoSI confidence regions satisfy instead

$$\liminf_{n\to\infty} P\left(\beta_{n,\hat{M}} \in \hat{R}_{n,\hat{M}}\right) \geq 1 - \alpha \tag{65}$$

for an arbitrary measurable $\hat{M}\colon \mathbb{R}^n \times \mathbb{R}^{n\times p} \to \mathcal{M}$, where $\mathcal{M}$ is the power set of $\{1, \ldots, p\}$.

Selective inference as simultaneous inference

Theorem

For any set of confidence regions $\{\hat{R}_{n,M} \colon M \in \mathcal{M}\}$ *and* $0 < \alpha < 1$,

$$P\left(\beta_{n,\hat{M}} \in \hat{R}_{n,\hat{M}}\right) \geq 1 - \alpha \text{ for all } \hat{M} \colon P(\hat{M} \in \mathcal{M}) = 1 \tag{66}$$

if and only if

$$P\left(\bigcap_{M \in \mathcal{M}} \left\{\beta_{n,M} \in \hat{R}_{n,M}\right\}\right) \geq 1 - \alpha. \tag{67}$$

$\rightarrow$ We can solve the selective inference problem (66) by solving the simultaneous inference problem (67).

For asymptotic validity of the confidence regions, require

$$\liminf_{n \to \infty} P\left(\bigcap_{M \in \mathcal{M}} \left\{\beta_{n,M} \in \hat{R}_{n,M}\right\}\right) \geq 1 - \alpha. \tag{68}$$

Selective inference as simultaneous inference

Proof.

For $M \in \mathcal{M}$, define the event $A_M = \left\{\beta_{n,M} \in \hat{R}_{n,M}\right\}$.

(67) $\Rightarrow$ (66): It is clear that

$$\bigcap_{M \in \mathcal{M}} A_M \subseteq A_{\hat{M}} \text{ for all } \hat{M}.$$

(66) $\Rightarrow$ (67): Let $\hat{M}$ be such that $\hat{M} \in \arg\min_{M \in \mathcal{M}} \mathbf{1}\{A_M\}$, so that $\mathbf{1}\{A_{\hat{M}}\} = \min_{M \in \mathcal{M}} \mathbf{1}\{A_M\}$. It follows that

$$\bigcap_{M \in \mathcal{M}} A_M = A_{\hat{M}}.$$

□

Selective inference as simultaneous inference

Worst possible selection rule:

The second part of the proof shows existence of an "adversarial selection rule" that picks a model if the corresponding parameter is not covered by the confidence region.

As in the fixed-X case, this selection rule is not actionable, as it depends on the population parameter.

Inherent high-dimensionality:

In view of the theorem, selective inference requires inference for a very large number of parameters $\beta_{n,M}$.

For instance, if $p < n$, there are $p2^{p-1}$ parameters in the various submodels.

UPoSI confidence regions

Recall the second-order arrays and their estimators:

$$\Sigma_n = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[X_i X_i^T\right], \qquad \hat{\Sigma}_n = \frac{1}{n}\sum_{i=1}^{n} X_i X_i^T \tag{69}$$

$$\Gamma_n = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[X_i Y_i^T\right], \qquad \hat{\Gamma}_n = \frac{1}{n}\sum_{i=1}^{n} X_i Y_i^T \tag{70}$$

and their submodel versions $\Sigma_n(M), \hat{\Sigma}_n(M), \Gamma_n(M), \hat{\Gamma}_n(M)$, and introduce the estimation errors

$$\mathcal{D}_n^{\Sigma} = \|\hat{\Sigma}_n - \Sigma_n\|_\infty = \max_{M\in\mathcal{M}_2} \|\hat{\Sigma}_n(M) - \Sigma_n(M)\|_\infty \tag{71}$$

$$\mathcal{D}_n^{\Gamma} = \|\hat{\Gamma}_n - \Gamma_n\|_\infty = \max_{M\in\mathcal{M}_1} \|\hat{\Gamma}_n(M) - \Gamma_n(M)\|_\infty, \tag{72}$$

where $\mathcal{M}_k = \{M\colon |M| \leq k\}$.

UPoSI confidence regions

The UPoSI confidence regions are

$$\hat{R}_{n,M} = \{\theta \in \mathbb{R}^{|M|} \colon \|\hat{\Sigma}_n\{\hat{\beta}_{n,M} - \theta\}\|_\infty \leq C_n^\Gamma(\alpha) + C_n^\Sigma(\alpha)\|\theta\|_1\} \tag{73}$$

$$\hat{R}^\dagger_{n,M} = \{\theta \in \mathbb{R}^{|M|} \colon \|\hat{\Sigma}_n\{\hat{\beta}_{n,M} - \theta\}\|_\infty \leq C_n^\Gamma(\alpha) + C_n^\Sigma(\alpha)\|\hat{\beta}_{n,M}\|_1\} \tag{74}$$

where $C_n^\Gamma(\alpha)$ and $C_n^\Sigma(\alpha)$ are bivariate joint upper α quantiles of $\mathcal{D}_n^\Gamma$ and $\mathcal{D}_n^\Sigma$:

$$P\left(\mathcal{D}_n^\Gamma \leq C_n^\Gamma(\alpha) \text{ and } \mathcal{D}_n^\Sigma \leq C_n^\Sigma(\alpha)\right) \geq 1 - \alpha. \tag{75}$$

<u>Note</u>:

- $\hat{R}_{n,M}$ has finite-sample guarantees and requires less assumptions, but it is more difficult to interpret and analyse.
- By contrast, $\hat{R}^\dagger_{n,M}$ provides asymptotic guarantees and requires additional assumptions, but they are more transparent.

UPoSI confidence regions

Some considerations:

- Bivariate quantiles, and therefore the confidence regions, are not unique. One may marginally increase one and decrease the other suitably, maintaining the bivariate coverage probability.
- The quantiles need to be estimated from the data. A bootstrap approach can be used to this end.
- Under mild conditions, $\max\{C_n^{\Gamma}(\alpha), C_n^{\Sigma}(\alpha)\} \to 0$ as $n \to \infty$.
- As opposed to fixed-X PoSI, there is no gain here from using sparse model spaces. This is because the estimation errors depend only on models of sizes 1 and 2.

UPoSI confidence regions

Theorem

The confidence regions $\{\hat{R}_{n,M} \colon M \in \mathcal{M}\}$, *where*

$$\hat{R}_{n,M} = \{\theta \in \mathbb{R}^{|M|} \colon \|\hat{\Sigma}_n\{\hat{\beta}_{n,M} - \theta\}\|_\infty \leq C_n^\Gamma(\alpha) + C_n^\Sigma(\alpha)\|\theta\|_1\}, \quad (76)$$

satisfy

$$P\left(\bigcap_{M \in \mathcal{M}} \left\{\beta_{n,M} \in \hat{R}_{n,M}\right\}\right) \geq 1 - \alpha. \quad (77)$$

An analogous result holds for the alternative regions $\hat{R}^\dagger_{n,M}$ under extra regularity conditions. Note:

- If $|M| > n$, $\hat{\Sigma}_n$ is singular, and $\hat{R}_{n,M}$ contains a nontrivial affine subspace of $\mathbb{R}^p$.
- Independence is not required (though it is needed for quantile estimation).

UPoSI confidence regions

Proof.

The proof is based on deterministic inequalities. Start from

$$\hat{\Sigma}_n(M)(\hat{\beta}_{n,M} - \beta_{n,M}) + (\hat{\Sigma}_n(M) - \Sigma_n(M))\beta_{n,M} = \hat{\Gamma}_n(M) - \Gamma_n(M).$$

The triangle inequality gives

$$\begin{aligned}
&\|\hat{\Sigma}_n(M)(\hat{\beta}_{n,M} - \beta_{n,M})\|_\infty \\
&\qquad \leq \|\hat{\Gamma}_n(M) - \Gamma_n(M)\|_\infty + \|(\hat{\Sigma}_n(M) - \Sigma_n(M))\beta_{n,M}\|_\infty \\
&\qquad \leq \|\hat{\Gamma}_n(M) - \Gamma_n(M)\|_\infty + \|\hat{\Sigma}_n(M) - \Sigma_n(M)\|_\infty \|\beta_{n,M}\|_1 \\
&\qquad \leq \|\hat{\Gamma}_n - \Gamma_n\|_\infty + \|\hat{\Sigma}_n - \Sigma_n\|_\infty \|\beta_{n,M}\|_1.
\end{aligned}$$

Therefore

$$\mathbb{P}\left(\bigcap_{M \in \mathcal{M}} \left\{ \|\hat{\Sigma}_n(M)(\hat{\beta}_{n,M} - \beta_{n,M})\|_\infty \leq \mathcal{D}_n^\Gamma + \mathcal{D}_n^\Sigma \|\beta_{n,M}\|_1 \right\} \right) = 1. \quad (78)$$

Substituting the errors by the quantiles gives the result. □

UPoSI confidence regions

Some important remarks:

- A bootstrap algorithm can be used to estimate the error quantiles such that the confidence sets are asymptotically valid under the quasi-exponential asymptotic regime $\log(p)^7 = o(n)$.
- The UPoSI confidence sets are not hyperrectangles, and therefore do not immediately imply marginal confidence intervals for each component, but there exists algorithms for constructing the smallest covering hyperrectangle for the sets.
- This method can be applied also in the fixed-X setting and can in fact produce smaller, though less interpretable, regions.

UPoSI confidence regions

Simulation study with fixed-X:[2]

Comparison of PoSI, UPoSI and UPoSI smallest hyperrectangle.

The data-generating model is $Y_i = X_i^T \beta + \varepsilon_i$, with $\beta = (0, \ldots, 0)^T$ and $\varepsilon \sim N(0, 1)$ independently, $n = 200$, $p = 15$.

Three covariate settings considered:

A. *Orthogonal design*: $\hat{\Sigma}_n = I_p$.

B. *Exchangeable design*: $\hat{\Sigma}_n = I_p - (p+2)^{-1} \mathbf{1}_p \mathbf{1}_p^T$.

C. *Worst-case design*:

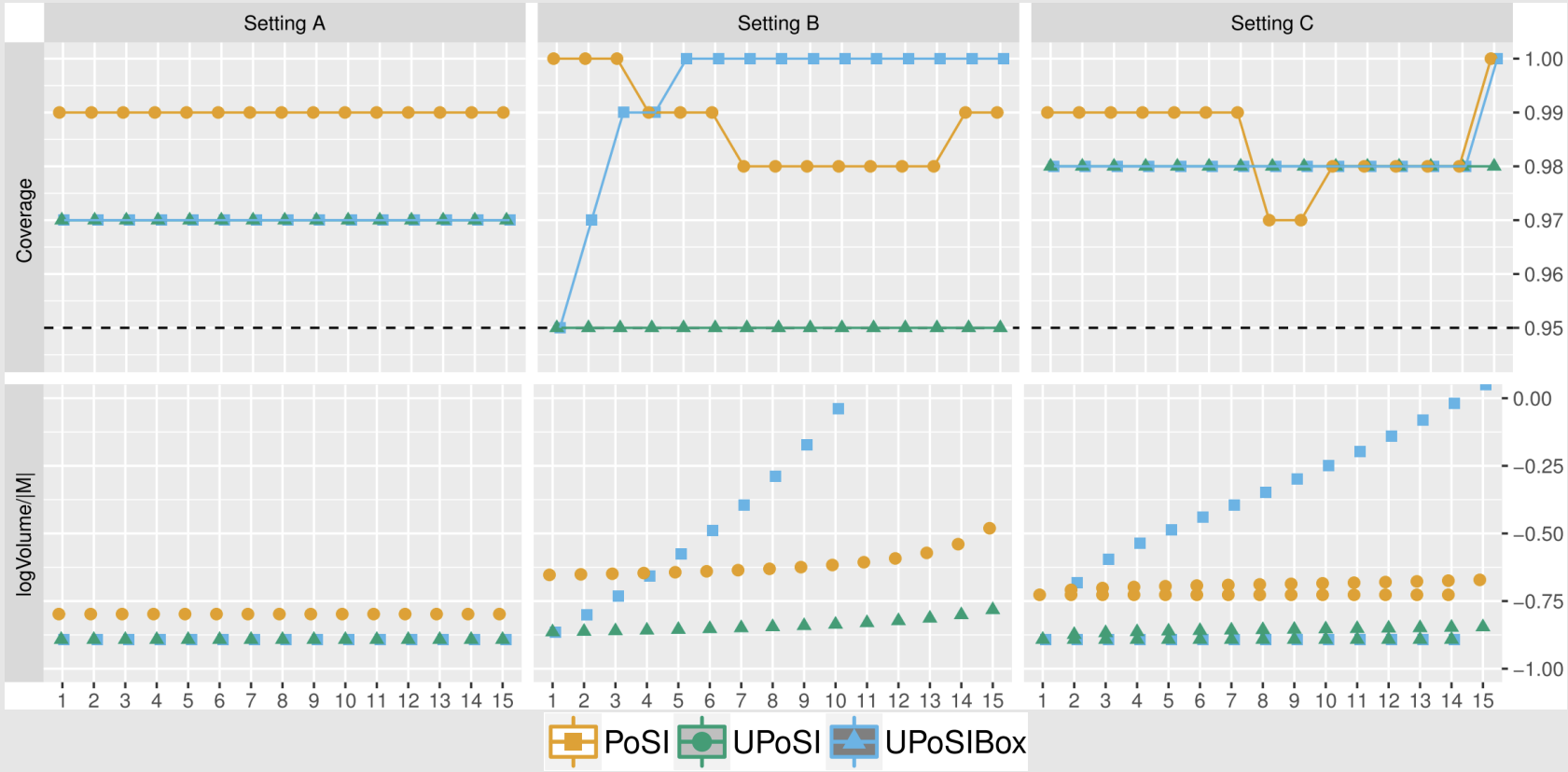
$$\hat{\Sigma}_n = \begin{bmatrix} I_{p-1} & c\mathbf{1}_{p-1} \\ \mathbf{0}_{p-1}^T & \sqrt{1/2} \end{bmatrix}. \tag{79}$$

Settings A and B are theoretically optimal for the original PoSI approach, while setting C leads to the largest PoSI constant.

[2] From Kuchibhotla, Brown, Buja, Cai, George, Zhao (2020). "Valid post-selection inference in model-free linear regression", *The Annals of Statistics*.

UPoSI confidence regions

Figure: Simulation results for 100 replications. Nominal coverage set at 95%. Plot shows simultaneous coverage and size of the confidence sets for different model sizes $1 \leq |M| \leq 15$. In setting C, models containing the last covariate produce significantly larger regions than those which do not.



Data splitting

An alternative, "obvious" approach to selective inference with a random design is sample splitting, whereby only a subset of the observations is used for model selection and the rest are used for inference.

Assume we have two subsets of IID observations

$$D_1 = \{(Y_i, X_i^T)^T : i = 1, \ldots, n\}, D_2 = \{(Y_i, X_i^T)^T : i = n+1, \ldots, 2n\}.$$

Since the data is IID, $\beta_{n,M}$ is independent of n, so we shall write β_M.

Furthermore, we restrict $\mathcal{M}$ to the set of M's such that the least squares problem has a unique solution, so that

$$\beta_M = \Sigma(M)^{-1}\Gamma(M),$$

where $\Sigma(M) = \mathbb{E}[X(M)X(M)^T]$ and $\Gamma(M) = \mathbb{E}[X(M)Y^T]$, and similarly for $\hat{\beta}_M$.

Data splitting

Clearly, if $\{\hat{R}_M \colon M \in \mathcal{M}\}$ are valid confidence sets for β_M for fixed M, they are also valid under selection for any selection rule $\hat{M}$, as

$$\mathbb{P}\left(\beta_{\hat{M}} \in \hat{R}_{\hat{M}}\right) = \sum_{M \in \mathcal{M}} \mathbb{P}\left(\beta_M \in \hat{R}_M \mid \hat{M} = M\right) \mathbb{P}\left(\hat{M} = M\right) \tag{80}$$

$$= \sum_{M \in \mathcal{M}} \mathbb{P}\left(\beta_M \in \hat{R}_M\right) \mathbb{P}\left(\hat{M} = M\right) \geq 1 - \alpha. \tag{81}$$

Trivially, they are also valid conditionally on the selected model, $\hat{M} = M$.

For a fixed M, standard inferential procedures can be used to construct asymptotic non-parametric confidence sets.

Data splitting: confidence sets based on asymptotic normality

Under suitable conditions, $\sqrt{n}\left\{\beta_M - \hat{\beta}_M\right\} \xrightarrow{d} N(0, V_M)$ for a positive definite matrix V_M.

The confidence sets based on a normal approximation are

$$\hat{R}_M = \left\{\beta \in \mathbb{R}^{|M|} \colon \|\beta - \hat{\beta}_M\|_\infty \leq \frac{t_\alpha}{\sqrt{n}}\right\}, \tag{82}$$

where t_α is such that

$$\mathbb{P}\left(\|\hat{V}_M^{1/2} Z\|_\infty \leq t_\alpha\right) = \alpha, \tag{83}$$

Z follows an $|M|$-dimensional standard Gaussian distribution independent of the data, and $\hat{V}_M$ is a plug-in estimator of V_M.

Under certain distributional assumptions and bounded model spaces $\mathcal{M}_n$ with $\max_{M \in \mathcal{M}_n} |M| = o(n^{1/5})$, we have, for all $\hat{M}$,

$$\liminf_{n \to \infty} \mathbb{P}(\beta_{\hat{M}} \in \hat{R}_M) \geq 1 - \alpha. \tag{84}$$

Data splitting: bootstrap confidence sets

Evaluation of $\hat{V}_M$ and computation of the quantile t_α turns out to be computationally demanding in some settings.

Alternatively, let t_α^* be the smallest positive number such that

$$\mathbb{P}\left(\sqrt{n}\|\hat{\beta}_M^* - \hat{\beta}_M\|_\infty \leq t_\alpha^* \mid D_2\right) \geq 1 - \alpha, \tag{85}$$

where $\hat{\beta}_M^*$ is a (standard) bootstrap copy of $\hat{\beta}_M$ using D_2.

The bootstrap confidence sets are defined as

$$\hat{R}_M^* = \left\{\beta \in \mathbb{R}^{|M|} \colon \|\beta - \hat{\beta}_M\|_\infty \leq \frac{t_\alpha^*}{\sqrt{n}}\right\}. \tag{86}$$

These confidence sets are asymptotically valid under similar conditions as the previous ones.

Data splitting

We have presented these constructions for the LS parameter β_M, but the framework can be applied to other type of inferential targets:

- The **Leave out covariate inference (LOCO)** parameters, which measure the importance of the selected covariates:

$$\gamma_j(M) = \mathbb{E}\left[|Y - \hat{\beta}_{M_j}^T X(M_j)| - |Y - \hat{\beta}_M^T X(M)|\right], \tag{87}$$

  where $\hat{\beta}_M$ is any estimator of β_M, and M_j and $\hat{\beta}_{M_j}$ are obtained by re-running model selection and estimation after removing the j-th covariate from the data.

- The **prediction parameter**, which measures how well the selected model will predict future observations:

$$\rho_M = \mathbb{E}\left[|Y - \hat{\beta}_M^T X(M)|\right]. \tag{88}$$

Data splitting

Concluding remarks:

- The main drawback is that it relies on the practitioner committing themselves to look only at a subset of the data during the selection stage. Thus, it does not protect against the bad practice of selecting the most convenient split and not acknowledging it.
- Ignoring some observations in the selection stage might even by inadvisable; e.g. if selection is the main goal or if one wants to study the stability of the selection procedure with respect to the split.
- It can sometimes have very little power in both stages (more powerful splitting strategies will be discussed later).
- The arbitrariness of the data split can be unsettling: two different statisticians using the same dataset but different splits can end up providing inference for different parameters.
- It requires identically distributed observations, while UPoSI does not.

Project

Objective: select an article from the following list and provide a **critical summary of it**, describing its main contributions and limitations and **relating it to the different frameworks of inference** presented in this course.

Project: proposed articles

- "Principled statistical inference in data science", Kuffner and Young (2018), *Statistical Data Science*.
- "Locally simultaneous inference", Zrnic and Fithian (2023), *arXiv:2212.09009*.
- "Hybrid confidence intervals for informative uniform asymptotic inference after model selection", McCloskey (2021), *arXiv:2011.12873*.
- "Valid confidence intervals for post-model-selection predictors", Bachoc, Leeb and Pötscher (2019), *The Annals of Statistics*.
- "Exact post-selection inference for sequential regression procedures", R. Tibshirani, Taylor, Lockhart and Tibshirani (2016), *Journal of the American Statistical Association*.
- "Black box selective inference via bootsrapping", Liu, Markovic and Taylor (2022), *arXiv:2203.14504*.
- "A note concerning a selection paradox of Dawid's", Senn (2008), *The American Statistician*.